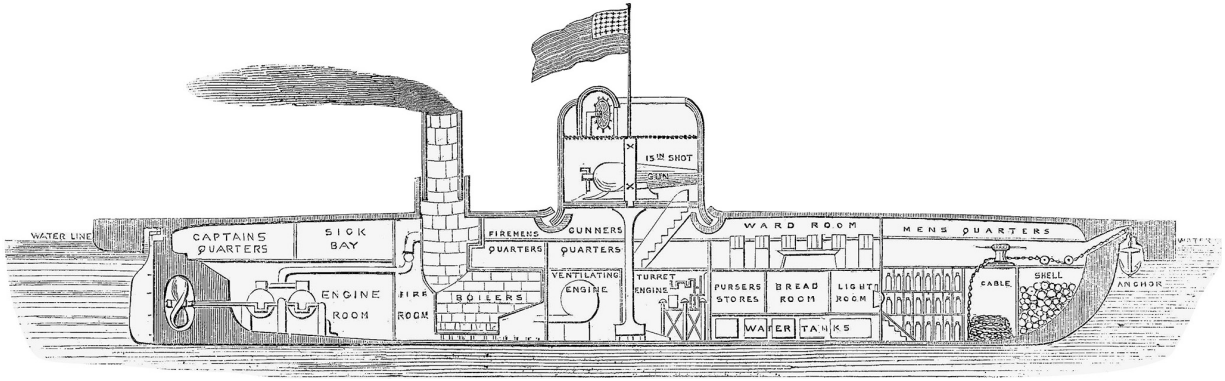




Cross-section of an ironclad

Most Union ironclads, including the *Weehawken* (above), were shallow-draft gunboats. Since none were very seaworthy, they were chiefly deployed in rivers and bays, and along the coast.

“... these **cheap, convenient, and formidable** devices ...”

UNION ADMIRAL JOHN A. DAHLGREN, AFTER THE *HUNLEY* SANK THE *HOUSATONIC*

Underwater attack

The use of spar torpedoes dramatically changed naval warfare, because they gave small ships the ability to damage or sink larger ones. Lookouts posted in the tops of Union vessels blockading Charleston harbor were always on the alert for wakes on the surface of the water from semi-submersibles.

In August 1863, a train pulled into Charleston carrying, at General Beauregard’s request, a flatcar with a very special freight. Unveiled was a 40-ft (12-m) submersible made from an old locomotive boiler with a tapered bow and stern added. Only 4ft (1.2m) in diameter, she was a tight fit for her crew of nine, one to steer and eight to crank her propeller. Though several practice dives proved fatal—one of them taking the life of its inventor, Horace L. Hunley—another crew volunteered, commanded by Lieutenant George Dixon. Christened *CSS Hunley*, her craft had a spar torpedo attached to her snout. When attacking, *Hunley* would stay just above